OOAIA

Lab Assignment 01

Class Specification

Implement a class `Matrix` with the following private members:

- `int rows` — number of rows
- `int columns` — number of columns
- `int** mat` — dynamically allocated 2D array

Constructor

You need to implement one specific constructor used by the driver code:

- `Matrix(int r, int c, int** matrix)`: Allocate memory for the class's internal matrix and **copy values** from the given input array.

Member Functions

1. `int trace()`
   Return the sum of diagonal elements ($A[i][i]$). If the matrix is not square, return `-1`.

2. `bool checkTriangular()`
   Return `true` if the matrix is either:
   - **Upper Triangular**: All elements below the main diagonal are zero.
   - **Lower Triangular**: All elements above the main diagonal are zero.

3. `void rotate90()`
   Rotate the matrix 90° clockwise.
   - You are **not allowed** to use an extra auxiliary matrix to store the result and assume that this function will only be tested on square matrices.
   - After rotation, print the updated matrix row-wise.

Input Format

- Line 1: N M (rows and columns)
- Next N lines: M integers each (matrix data)
- Next line: Q (number of queries)
- Next Q lines: query ID (1 to 3)

Query Mapping

ID	Function	Output
1	`trace()`	Integer result
2	`checkTriangular()`	true / false
3	`rotate90()`	Rotated matrix

Sample Input

```
3 3
1 2 3
4 5 6
7 8 9
3
1
2
3
```

Sample Output

```
15
false
7 4 1
8 5 2
9 6 3
```